Adaptive Graph Transform Clustering via Rate-Distortion Optimization for Point Cloud Attribute Compression

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Introduction

Point clouds (PC) represent 3D geometry and attributes with high fidelity but massive data footprints [8]. This work develops **Point Cloud Attribute-Dependent Compression (PCADC)**, which adapts the graph topology and transform selection to local signal characteristics. We solve the *signaling tax* problem by leveraging overhead dilution in large block volumes.

Methodology: Partitioning and Topology

The PC is partitioned using a **Morton Path** to preserve spatial locality [2]. We introduce the **Attribute-Dependent Graph** ($\mathcal{G}_a$), which adapts the structural graph $\mathcal{G}_s$ by adding self-loops to nodes identified as "sinks" in the luminance gradient field.

$$L_a = L_s + \mathbf{W}_{sl} \quad \text{where } \mathbf{W}_{sl} = \text{diag}(w_i \cdot \mathbb{1}(i \in \text{Sinks})) \tag{1}$$

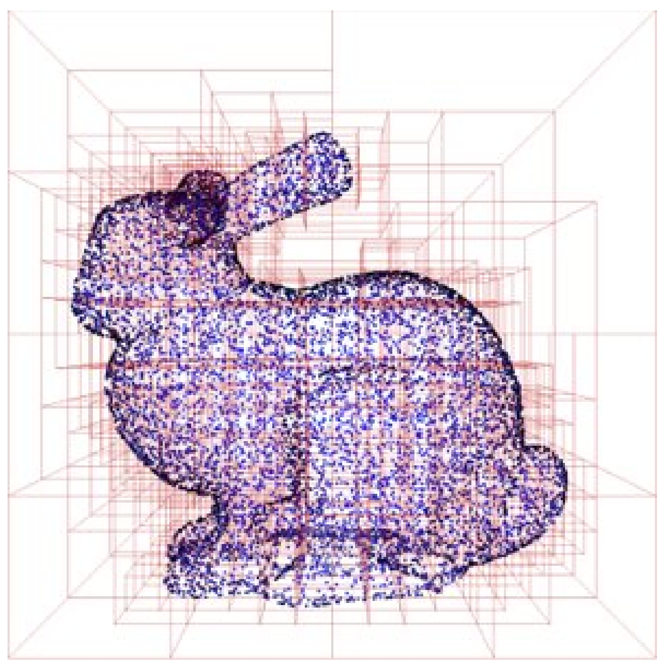

Figure 1. Morton partitioning.

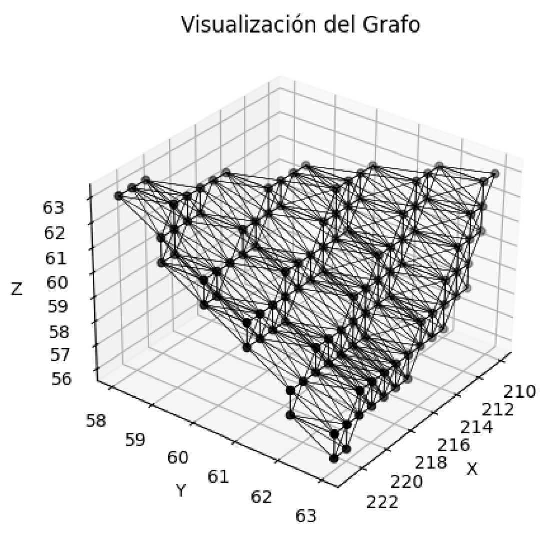

Figure 2. Adaptive Topology.

Learning the Dictionary: RD-Optimized Clustering

We learn a codebook of K transforms to minimize the Lagrangian cost $J = D + \lambda R$ [3].

- **Exploration:** *Simulated Annealing* enables stochastic assignments to avoid local minima.
- **Refinement:** *Differential Evolution* refines transform parameters $(\vec{slope}, w_{sl}, \tau)$ to minimize the collective cost [5].

Information Theory: The Signaling Tax

A core finding is the **Overhead Trap**. For small blocks (B4), signaling which transform to use costs ∼0.2 BPV, exceeding transform gains. **The Dilution Principle:** Per-point overhead drops with block volume B^3:

$$\text{BPV}_{\text{overhead}} = \frac{H(L_i|L_{i-1})}{B^3} \quad [7, 6] \tag{2}$$

True compression requires $B \geq 16$ to dilute the tax.

Geometric Complexity and Vanishing Clusters

Analysis of *Active Clusters* reveals a "Structural Ceiling." B4 blocks hit a limit of ∼8 distinct modes regardless of capacity [4]. B32 blocks retain higher entropy, justifying more complex codebooks.

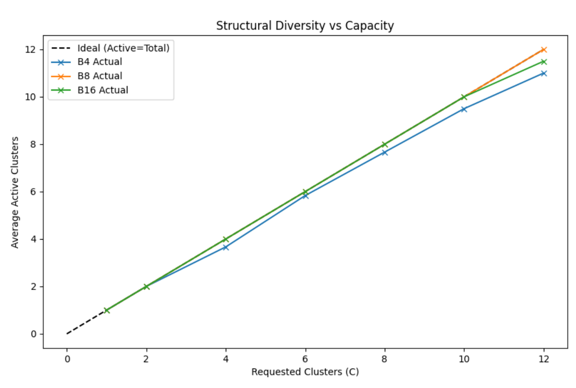

Figure 3. Utilization vs Capacity.

The Capacity Elbow: Diminishing Returns

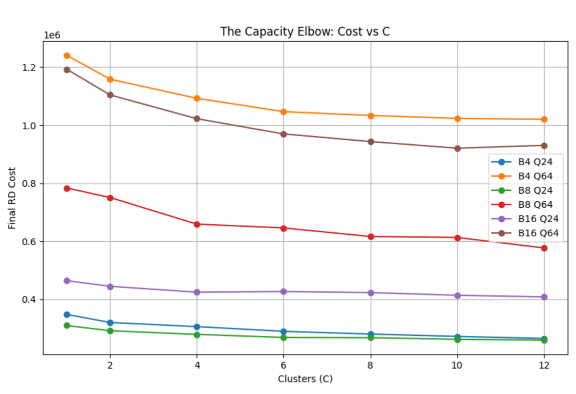

Figure 4. Cost vs Clusters.

The inflection point identifies the optimal K. For B32, $K = 2$ captures the primary signal trends, while $K > 4$ leads to saturation where overhead growth negates residual reduction.

Spatial Regularization and CABAC

To reach true compression, we implemented **Spatial Regularization**, adding a penalty β for cluster-switching. This forces label "runs" efficiently compressed using **CABAC-Proxy** (1st-order conditional entropy), exploiting spatial correlation [1].

Rate-Distortion Frontier: Breakthrough at B32

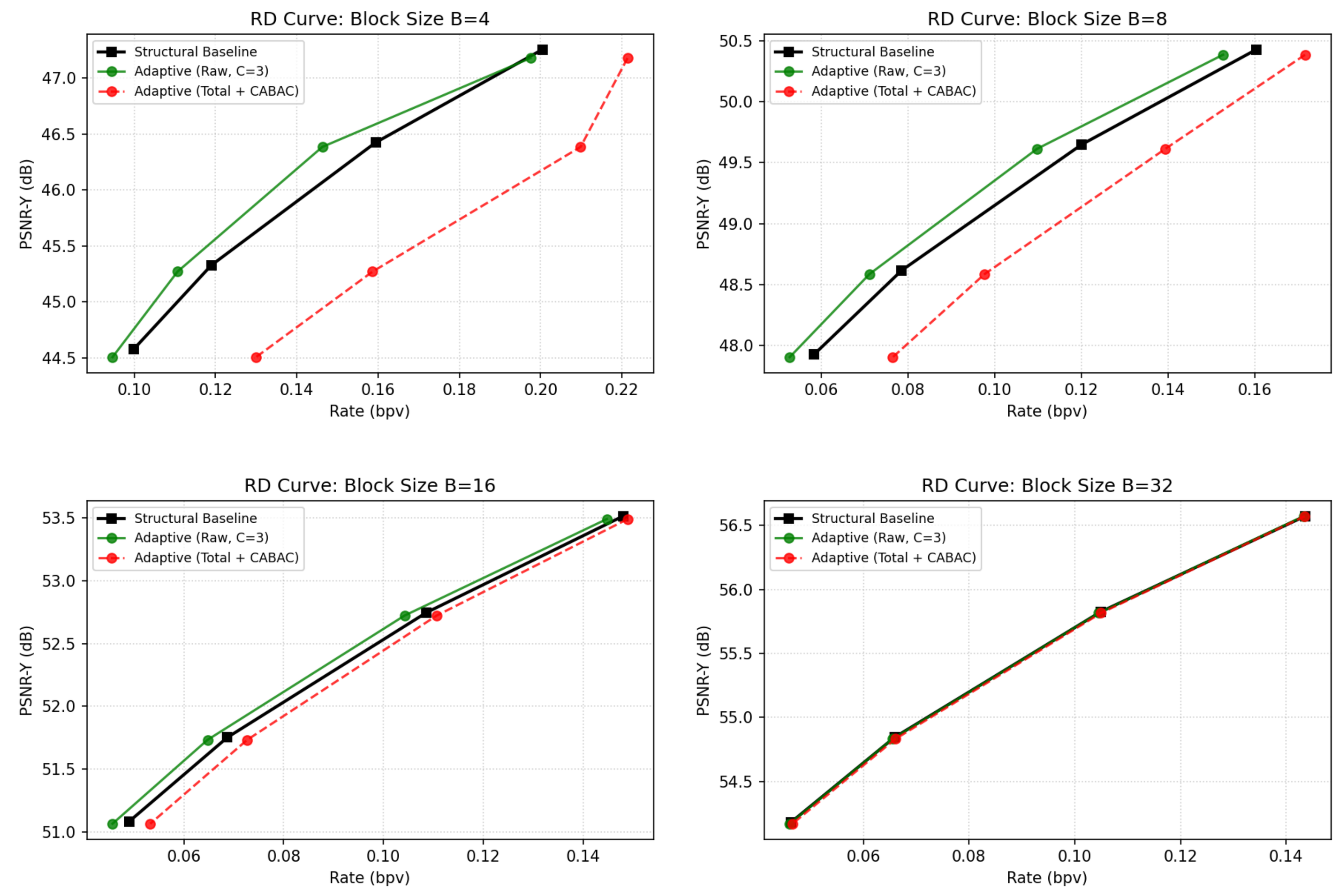

Figure 5. RD Curves (PSNR-Y vs bpv). Dotted Red: Total Rate (Data + CABAC). In **B32**, the gains finally outweigh the tax, achieving net reduction.

Config ($K = 4$)	Overhead (bpv)	Raw saving	Net Gain
Small Detail (B4)	0.0239	-1.4%	+10.47%
Medium Scale (B8)	0.0189	-4.8%	+7.02%
Large Scale (B16)	0.0042	-2.2%	+0.61%
Compression (B32)	**0.0002**	**-0.2%**	**-0.03%**

Table 1. Net performance using context-adaptive signaling proxy.

Conclusions

True compression in PCADC requires **overhead dilution** and **spatial smoothing**. While B4/B8 allow superior local adaptation, their signaling cost is prohibitive. The optimal strategy is **B32 with $K = 2$ and high β**, achieving the project's first true compression breakthrough.

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